



DATABASE EXTREME
PRODUCT OVERVIEW · V1.2.0

A DATABASE FOR MULTI-TENANT PRODUCTS

Every customer gets a database.

Durable application state and vector memory, organized around the tenant boundary and operated from one self-hosted product.

01

Tenant-first

02

KV + vectors

03

Self-hosted

dbx · product experience



Tenant separation is easy to get wrong.



When customer boundaries live in application conventions, every workflow must remember them.

01

Keys carry identity

Tenant IDs are repeated in key prefixes and query paths.

02

Access checks repeat

Each service must apply the right customer scope.

03

Lifecycle fragments

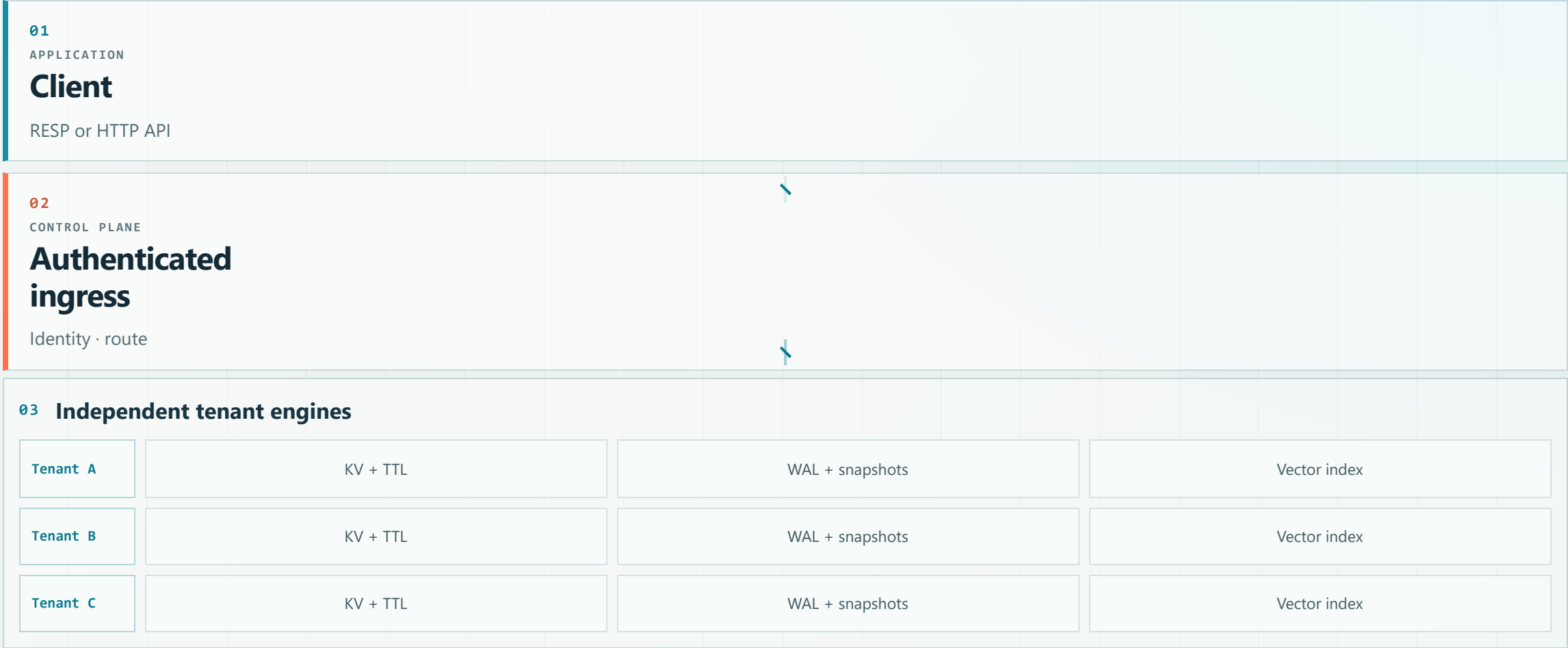
Backup and deletion span scripts and storage conventions.

DBX DESIGN CHOICE

Make the tenant the database boundary.

One ingress. *Independent tenant engines.*

Authenticate once, then route each request to its tenant-scoped data engine.



Working state and semantic recall, inside one tenant boundary.

01 STATE

Key-value store

Application state that is quick to read, update, and expire.

- ↳ RESP-compatible strings
- ↳ TTL, counters, and persistence
- ↳ Redis client libraries: Python, Node, Go

```
SET session:42 "active" EX 1800
```

ONE ENGINE +

02 RECALL

Vector search

Embeddings come from the caller. DBX does not run embedding, image, or multimodal models. Caller-provided embeddings indexed for approximate nearest-neighbor retrieval.



- ↳ Sharded HNSW + SQ8 storage
- ↳ VSEARCH and VSIM

DBX • Database product deck

Explore vector retrieval before wiring an app.

Try a tenant index, adjust search controls, and inspect ranked results from one workspace.

VSEARCH

VSIM · VFUSE

DBX
CONTROL PLANE

Tenant > searchlab > Vector searchlab

Search

TENANT

- Overview
- Data Explorer
- Vector Playground
- Tenant keys
- Console

RUNTIME

- Hardware
- Storage
- Network
- Hosting

CONTROL PLANE

- Settings
- Security
- Replication

Vector Playground

Query this tenant's vector index the same way an application would.

● RUNNING
HTTP :8081
RESP :6401
Keys 6

Scores are cosine similarity × 100. VSIM reads the stored row and excludes that id. DBX does not run a model.

Similarity search

Semantic
Similar-to-Id
Multimodal

INDEX NAME
test_index

STORED VECTOR ID

Results

#1
36.69

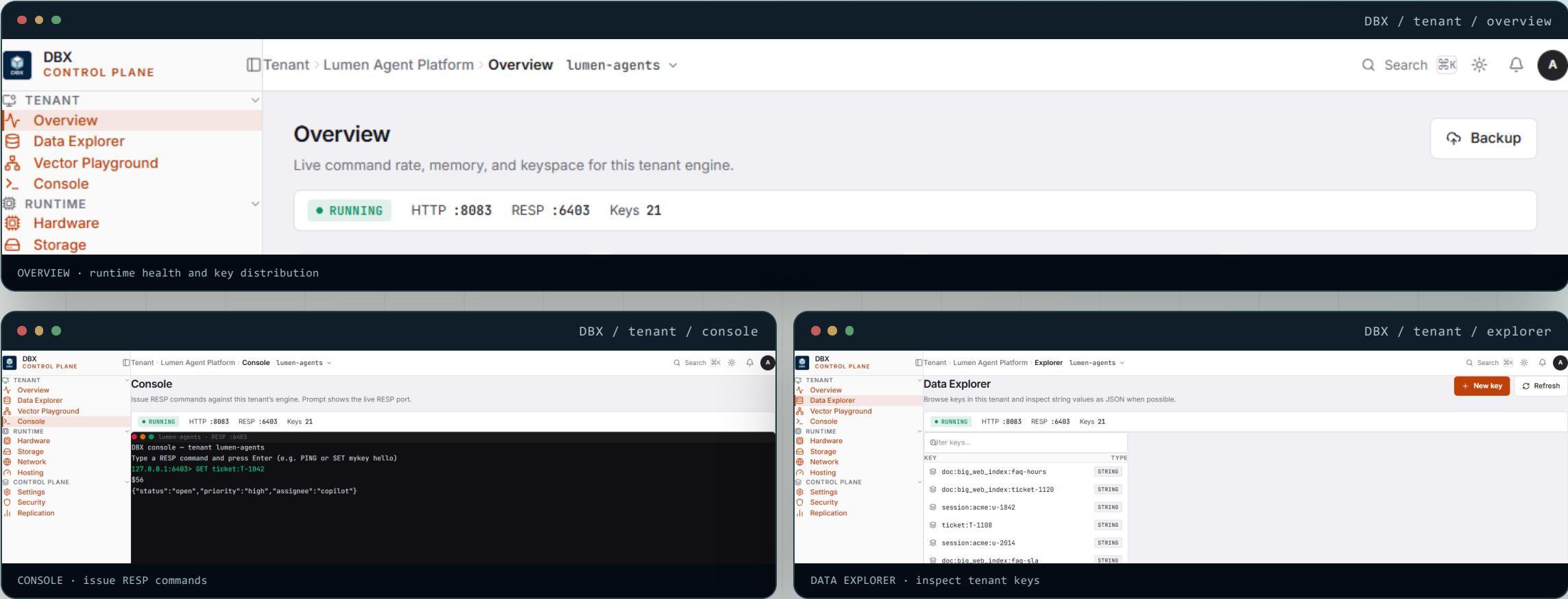
payroll handbook and benefits

ID: bravo

#2
18.88

Vector Playground · query controls and ranked neighbors in the DBX dashboard

From live health to raw data, without leaving the product.



Tenant separation reaches the worker and its data lifecycle.

14-17 MiB idle RSS
per strict worker

**Clear separation.
Honest storage boundary.**

Scoped roles, quotas, usage APIs, and tenant lifecycle actions are managed through the control plane.

AT-REST DETAIL

Customer-level operations are built into the data model.

01

Recover

WAL + RECOVERY

Checksummed WAL and checkpoints restore tenant data after restart.

02

Back up

TENANT ARCHIVE

Versioned tenant archives carry SHA-256 checksums across KV and vector state.

03

Change safely

SHADOW MIGRATION

Build and promote a replacement vector index while the current index remains searchable.

04

Investigate

TIME-TRAVEL

Search retained vector state as of a timestamp for retrieval analysis.

05

Replicate

ASYNC WAL

Optional WAL replicas receive changes asynchronously; primary writes acknowledge locally.

06

Observe

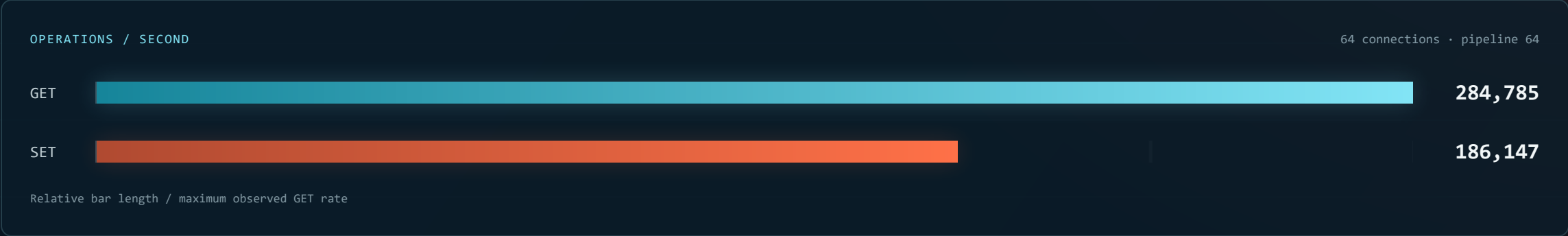
OPERATOR UI

Console, explorer, playground, tenant keys, and runtime views ship with the deployment.



Fast key-value operations on a measured host.

CERTIFICATION RUN
27 AUG 2026



DBX / KV

128k

operations per workload

1

tenant · no eviction

Every second

WAL sync setting

Windows 10 · 18 logical processors · 15.6 GiB RAM

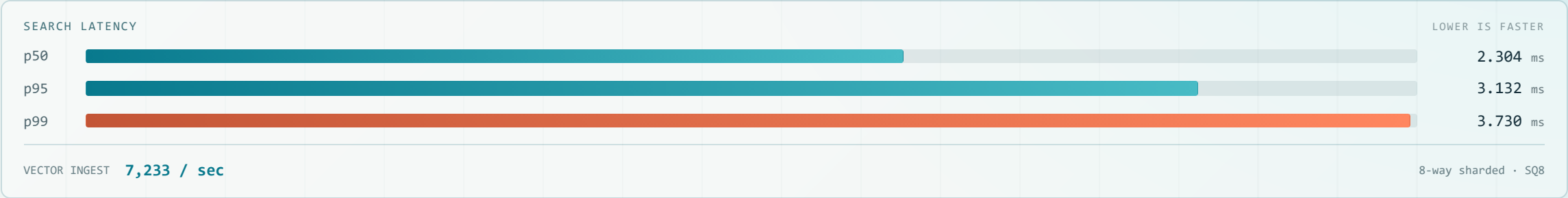
Loopback RESP · 27 Aug 2026

Measured on the Windows 10 certification host; this run does not certify Linux performance. Results vary with host, durability settings, workload, and client. Methodology: [scripts/benchmarks/performance_analysis.md](#).

Latency with recall, on the same slide.

100,000 vectors × 128 dimensions · float32 brute-force recall baseline

RECALL@10 MEAN
0.920 p05 · 0.800



27 Aug 2026 · Windows 10 · 18 logical processors · 15.6 GiB RAM. Linux CI reruns vector gates; this Windows host does not certify Linux performance. Methodology: [scripts/benchmarks/performance_analysis.md](#).

Built for bounded, per-customer working sets.

A self-hosted database with tenant-scoped application state, vector search, and an included operator dashboard.

100

tenants / node

100k

vectors / tenant

CURRENT BOUNDARIES

Single-node v1 · no cluster or cross-node sharding · no SQL joins · no billion-vector claim.

dbx / product / features

DBX

10-11-11-11

Product

Docs

Updates

Pricing

Security

Contact

Get started

Search

Clear

HOW / FEATURES

0-100% SLIDESHOW / ONE EXPERIENCE

Five claims. One lifecycle.

The tenant is a first-class object. Everything else — RESP, HNSW, the dashboard — is in service of that.

01 Tenant is a first-class object

RESP / api/permissions creates a first isolated engine: its own data directory, write-ahead log, HNSW state, snapshot lineage, and process-level state isolation is structural, not a naming convention. Provision, back up, restore, and delete operate on exactly one customer. There is no cross-tenant scan and no shared keyspace to sweep.

02 State and memory in one engine

KV with TTL (session, scratchpad, rate counters) and HNSW vectors (semantic recall) live in the same process, behind one connection, in one data directory, and inside one backup archive. An agent's session and its recall cannot drift apart — there is no second system to drift from. Based on that tenant is three commands: `rescan` (query vector), `vsim` (neighbors of a stored id; self excluded), and `vsim` (date-Rule across nearest `vsim` graphs). Optional `mid_score` and per-query `at` stay on the same wire. Embeddings stay with the caller. CTR does not run CTR on MIMIC4. Harvest limits the parallel `vsim` against rate counters KV only. The tenant backup archive is the point-in-time and `checkpointed` `vsim` plus vector files, bound by state that recovery verifies before `vsim` restore. Do not read that in: vectors live inside the `vsim`. `vsim` is a weighted sum of per-space cosine, not a multistage model. Harvest spaces are extra.

DBX product overview · self-hosted database and operator workflow



DATABASE EXTREME

DBX · V1.2.0

A DATABASE BOUNDARY FOR EVERY CUSTOMER

One customer. One database.

Durable state, vector memory, and tenant-level operations in one self-hosted database.

dbx-database-extreme.com

github.com/vanshjain-0702/DBX-Database-Extreme

